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In situ absorption sensor for NO in combustion gases with a 5.2 μm quantum-cascade laser

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Abstract

A new mid-infrared absorption sensor for *in situ* detection of nitric oxide in combustion exhaust gases has been developed and demonstrated for temperatures up to 700 K. A novel external-cavity quantum-cascade diode laser, which can be wavelength-tuned over the R-branch of the fundamental absorption band near 5.2 μm, was utilized, enabling critical evaluation of the interference absorption by H_2O in combustion exhaust gases. The water vapor absorption spectrum was measured over the range 1880 cm^{-1} to 1951 cm^{-1} at 633 K (680 F), typical of conditions in the economizer region of coal-fired power plants. Based on the data for water vapor interference and laser performance, four candidate fundamental-band transitions (R10.5 (1/2, 3/2) and R15.5 (1/2, 3/2)) were selected as optimum and subsequently investigated in detail. A laboratory combustion exhaust rig with a 1.79 m constant-temperature line-of-sight path was then used to validate the sensor for NO concentrations between 20 and 95 ppm at 600 K; an NO detection limit of <60 ppb m $(Hz)^{-1/2}$ was obtained. The sensor was subsequently applied in successful real-time measurements of NO across a 3 m path in the exhaust of a pulverized-coal-fired power plant, including capture of a 30% transient change in NO concentration at the economizer exit that occurred during the shutdown and recovery of the SNCR NO_x control system. The sensor shows excellent promise for monitoring NO in practical combustion exhausts of coal-fired power plants at temperatures up to at least 700 K.
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Keywords: Nitric oxide; Combustion exhaust; *In situ*; Quantum-cascade laser; Absorption spectroscopy

1. Introduction

Nitric oxide (NO) is an important atmospheric constituent of the oxides of nitrogen (NO_x), and the majority of this pollutant is formed from the combustion of fossil fuels. NO contributes to ground-level ozone, acid rain, and a variety of adverse human health effects, which has led to increasingly stringent regulatory mandates on the emission of NO_x [1–3]. As a result, the electric-power industry has a growing need to develop reliable control systems to suppress the release of NO_x effluent from combustion-driven generation of electricity. Accurate and reliable real-time sensors are an important part of any control system, and measurements from a variety of locations in the power plant, ranging from the combustor to the atmospheric exit of the exhaust stack, could facilitate the needed NO_x control. The residence times in such power plants can be large, and thus,

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as NO is monitored closer to the combustion zone, the response time of a control system can become shorter.

Traditional extractive sampling methods suffer from long response time and have the potential for secondary chemistry in the sampling probes. There is thus strong motivation for developing sensors for *in situ* monitoring of NO in combustion exhaust streams. Optical absorption methods have the potential for fast, sensitive, accurate and species-selective*in situ* measurements in gas streams over a wide variety of conditions (e.g. temperature, pressure, gas composition) even in the presence of particulate such as fly ash, and owing to their high spectral intensity and potential for accessing strong absorption transitions, laser-based schemes show particular promise for practical applications, including power plants [4–11]. NO detection using laser absorption spectroscopy in the infrared has been one of the strategies extensively studied over the past few decades; see Fig. 1 for the NO absorption spectrum over the range 1–6 μm for a temperature of 600 K. Work has been reported using transitions in three different vibrational bands, the second overtone band ($\Delta v = 3$) near 1.8 μm [12,13], the first overtone band ($\Delta v = 2$) near 2.7 μm [14], and the fundamental band ($\Delta v = 1$) near 5.2 μm [15–17].

The absorption line strengths of CO, CO_2 and H_2O from the HITRAN database [18] are shown in Fig. 1 to illustrate the potential for interference absorption in hot combustion exhaust gases. The fundamental band holds the most promising candidate transitions in terms of their strong absorption (about 100-times stronger line strengths than in the first overtone band near 2.65 μm) and relatively weaker interference transitions of the other combustion species.

Refs. [15–17] illustrate the history of NO detection using transitions in the fundamental band. The first NO detection by tunable-diode-laser absorption in flames using fundamental-band transitions was conducted over twenty-five years ago by Falcone et al. at Stanford University [15]; unfortunately, the only lasers near 5.2 μm at that time were low-power, multimode, cryogenically cooled lead-salt devices. More recently, NO detection in the mid-infrared has benefitted from the improvement of room-temperature (thermoelectrically cooled) quantum-cascade (QC) laser technology, now available both in pulsed and continuous-wave devices. McManus et al. evaluated field instruments for atmospheric air quality that included absorption detection of NO at 1900 cm^{-1} using both pulsed and continuous-wave (CW) QC lasers [16]. With gas extraction at room temperature into an astigmatic multi-pass optical cell, detection precisions of 0.5 ppb $(Hz)^{-1/2}$ and 0.1 ppb $(Hz)^{-1/2}$ were reported, and long-term reliable use of QC laser technology was also demonstrated. Wysocki et al. [17] reported a 1σ noise-equivalent sensitivity of 100 ppb m $(Hz)^{-1/2}$ at room temperature and 200 ppb m $(Hz)^{-1/2}$ at 675 F (630 K) measured with a variety of gas mixtures in a heated cell, to illustrate the potential for a mid-IR absorption sensor for *in situ* sensing of NO in combustion exhaust gases.

Here we discuss the extension of QC-laser-based mid-IR absorption of NO to *in situ* detection in combustion exhausts. We first examine the fundamental band of NO and identify two pairs of promising transitions (R10.5 (1/2, 3/2)

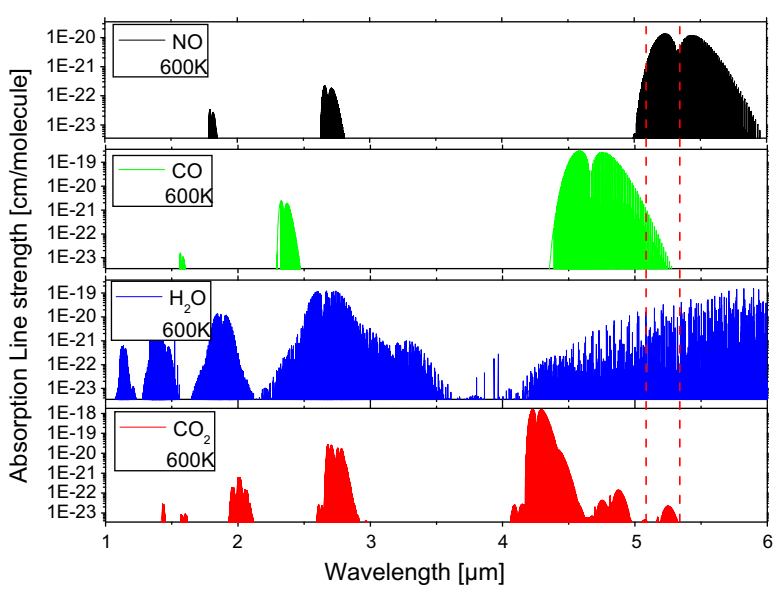

Fig. 1. Absorption line strengths in mid-IR region for NO, CO, H_2O and CO_2 at 600 K from HITRAN [18]. Transitions with line strengths less than 3.5×10^{-24} cm/molecule are not shown.

doublet and R15.5 (1/2, 3/2) doublet) and evaluate their performance in a laboratory combustion exhaust rig. Unlike optical detection of trace species at ambient (room temperature and pressure) conditions, *in situ* detection in elevated temperature combustion exhaust presents interference problems for optical absorption measurements. In Fig. 1, note that the vibrational bands of NO overlap with bands from combustion product water vapor (H_2O) and carbon dioxide (CO_2). In particular, interference from hot water vapor presents a great challenge due to its large mole fraction (~10%) in combustion exhaust, compared with the ppm level NO, and the presence of transitions of hot H_2O throughout the near- and mid-IR region. Therefore, a careful study of H_2O absorption spectra versus gas temperature is a critical step in establishing the sensor strategy and in evaluating the sensitivity of NO detection.

Very recently (December 2009), we were able to demonstrate 5.2 μm QC-laser-based NO absorption sensing in the exhaust of a pulverized-coal-fired power plant. The measurements were conducted at the exit of the economizer region, where the gas temperature was ~624 K (663 F), located downstream of a selective non-catalytic reduction (SNCR) system for NO using injection of dilute urea solution. The flow velocity in the combustor is sufficiently fast that this location is only a few seconds delayed from the intake into the firebox. Preliminary results of these field measurements are reported to further demonstrate the practical potential of a QC-laser-based NO absorption sensor in the mid-infrared.

2. Fundamental theory of laser absorption spectroscopy

Only a brief discussion of the fundamentals of absorption spectroscopy is reproduced here to define units and terms, since the theory is well understood and readily available in the literature [4]. According to Beer's Law, the fractional transmission of monochromatic light at frequency v through an absorbing gas media i is given by

$$\tau(v) = \left.\frac{I}{I_0}\right|_v = \exp\left(-SP\chi_i\phi_v L\right) \tag{1}$$

where I and I_0 are the transmitted and incident light intensity, respectively, S the line strength of a particular absorption line, P the total pressure, χ_i the mole fraction of i, ϕ_v the lineshape function, and L the optical path length. The dimensionless product $\alpha = SP\chi_i\phi_v L$ is defined as absorbance, and the lineshape function ϕ_v is normalized to have unit area across the line. Note that the line strength S is temperature-dependent, and can be calculated from the partition function and lower-state energy of the transition. Therefore, the integrated absorbance area

$$A = \int_{\text{line}} \alpha dv = SP\chi_i L \tag{2}$$

can be used to obtain time-resolved information on the concentration of NO, in the usual cases where S, P, and L are known or can be reasonably estimated.

3. Experimental setup for high temperature combustion exhaust measurements

A custom, CW external-cavity quantum-cascade laser (EC-QCL) was acquired from Daylight Solutions to access the R-branch of the fundamental band of NO. Measurements of water vapor absorption versus temperature were first conducted to select optimum transitions for NO detection in combustion gas at elevated temperature. The laser could be tuned over the range of 1880 cm^{-1} to 1951 cm^{-1} (5.1256 μm ~ 5.3191 μm) at a thermoelectrically-cooled chip temperature of 16 °C, and within this range a mode-hop-free region of over 40 cm^{-1} was available. The laser is single-mode, with a linewidth specified to be <0.001 cm^{-1}, qualifying it as a monochromatic light source for these spectroscopic measurements. The laser produces a free-space optical beam with power in excess of 40 mW, more than 50 times the typical power from lead-salt diode lasers in this wavelength region. This power should be sufficient for *in situ* long-path measurements in a harsh environment. Characterization of the spatial-mode quality of the beam found the divergence to be well-characterized by a non-astigmatic elliptical Gaussian profile (~1.5 mrad by 4 mrad) and a waist size of <0.5 mm.

The combustion exhaust test rig at Stanford consisted of a premixed flame and a temperature-controlled exhaust duct, illustrated in Fig. 2. A premixed flat flame burner (McKenna) supports an ethylene flame with air or a gas mixture of 21% O_2 and 79% argon, which prohibits the formation of NO in the flame. Complete combustion of a stoichiometric mixture has nominal H_2O and CO_2 mole fractions of 13%, providing the laboratory test environment needed to simulate coal-combustion exhaust gases.

The exhaust duct had a uniform temperature distribution along its length controlled with a heating manifold up to 1000 K (1340 F). The duct had an internal diameter of 59.5 mm, with 50.8 mm-diameter wedged CaF_2 windows at each end of the duct, providing a 1.79 m optical path. Controlled quantities of trace species such as NO and CO were added to the flow downstream of the burner, and gas mixtures in the duct were sampled at both ends of the optical path to provide a reference measurement of concentrations via traditional extractive sampling methods (O_2, CO, CO_2, via NDIR, and NO_x via chemiluminescence). The

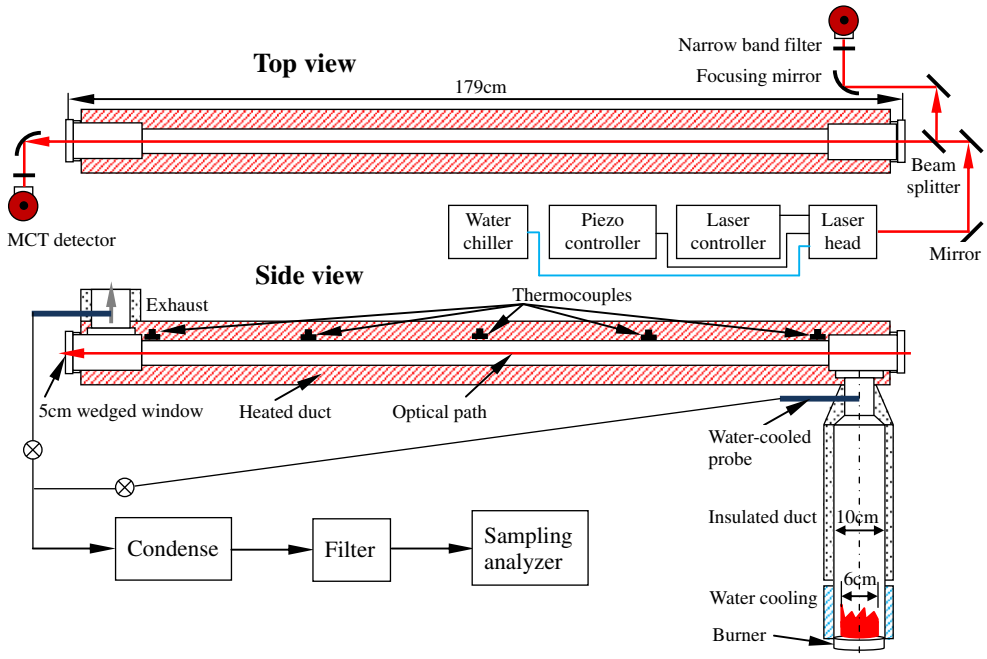

Fig. 2. Combustion exhaust rig showing laboratory burner, constant-temperature exhaust duct, optical access and gas-sampling ports.

top panel of Fig. 2 shows the optical setup of the experiment, where common-mode rejection was used to improve signal-to-noise ratio (SNR) with wavelength-scanned direct absorption. A ZnS beam splitter was used to split the laser beam into two arms to be received by a pair of matched liquid-nitrogen–cooled mercury–cadmium–telluride (MCT) detectors (future work will use thermoelectrically-cooled detectors); one beam passes through the absorbing gas media, while the other propagates through ambient air to serve as the reference beam for common-mode rejection.

4. Broad-spectral-range survey of absorption spectra and NO line selection

The wavelength scanning range of the external-cavity laser accessed the R-branch (R5.5–R24.5) in the fundamental vibrational band of NO near 5.2 μm. By rotating the grating inside the laser head, the laser could scan over the entire range in less than 5 s. Figure 3 shows the transmitted laser intensity through a 17.9 cm-long cell filled with 50 Torr of 1% NO in room temperature nitrogen in the upper trace and the analyzed absorption spectrum in the lower trace. Typical of external-cavity-laser architecture, as the laser-wavelength was grating-tuned the output intensity was modulated with constructive and destructive interference on the output coupler of the cavity. This produced an "etalon" interference pattern in the output intensity of approximately ±5% for this laser. The intensity pattern was used to provide an accurate laser-wavelength scale, via calibration with a wavemeter (Bristol 621 Series). The overall intensity variation with wavelength (a factor of three) and the etalon oscillation were accounted for using common-mode rejection; the result is illustrated in Fig. 3. The resulting absorption spectrum in the lower trace of Fig. 3 shows baseline SNR with a single-scan noise-equivalent absorbance of 6×10^{-3} and reveals the effectiveness of the common-mode rejection scheme, which will also be used in temperature-dependent measurements of water vapor interference in NO detection.

The data in Fig. 3 include strong absorption transitions of water vapor in the ambient laboratory air ($\chi_{H_2O} \sim 1\%$) near 1910, 1918, 1924 and 1943 cm^{-1}. This interference becomes a greater concern at elevated temperatures where the water vapor transitions with larger internal-energy become significant, and in combustion gases where the water vapor mole fraction is of order 10% and the NO is a trace (~100 ppm) constituent. In such circumstances, a weak, high-internal-energy water vapor transition can produce significant absorption interference. An accurate database of high-internal-energy water vapor transitions is not readily available in the literature. We believe the measurements reported here are

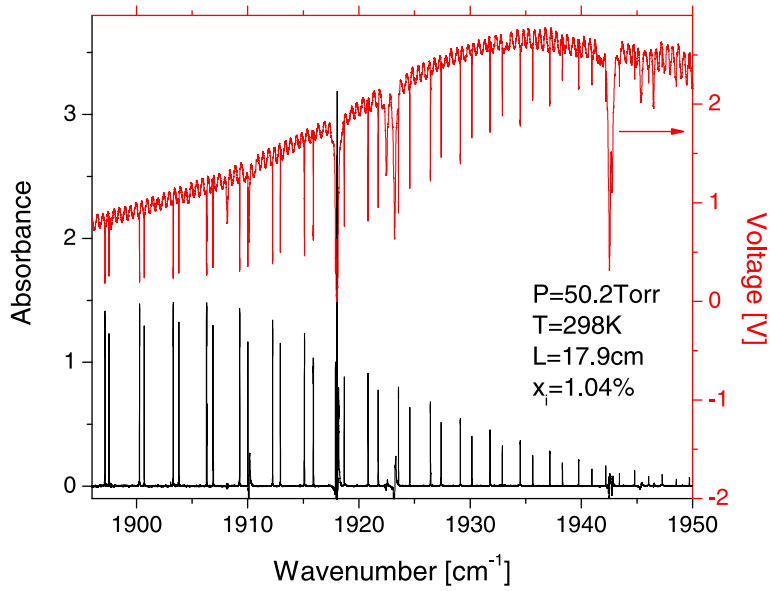

Fig. 3. Upper trace, laser transmission through a room temperature cell with 50 Torr of 1% NO in N_2; Lower trace: NO absorption spectrum with common-mode correction to the laser power and an etalon-based laser-wavelength scale.

the first sensitive measurements of water vapor absorption at elevated temperature in this wavelength region. These measurements are crucial to identifying the optimum "windows" in the H_2O absorption spectrum where sensitive NO measurements can be made, and in quantifying the interference-limited NO detection limit in combustion exhaust at elevated temperature. These gaps or "windows" in the water vapor absorption are temperature-dependent; therefore the optimum transitions for NO detection may vary with gas temperature.

Figure 4 compares the measurement of water vapor absorption spectrum at $T = 633$ K with HITRAN predictions. Only the wavelength region of 1897 cm^{-1} to 1935 cm^{-1} is shown for graphic clarity. The data were acquired in a fuel-lean ethylene flame to minimize NO production. To illustrate the potential for NO measurements, HITRAN predictions for 500 ppm of NO are also shown in the figure. Except for the water transition near 1929 cm^{-1}, the data of Fig. 4 show that HITRAN does an excellent job predicting the strong water absorption at 633 K. We find over a dozen possible choices of absorption lines (as indicated by the arrows in the graph) for detection of hundred-ppm-level NO, a value of NO typical for coal-fired combustion exhaust streams internal to a power plant. As temperature is increased above this level, some of the windows close up due to increasing water absorption, so that a number of these lines were rejected because of their proximity to water transitions (numbers 3, 4, 7, 10–13 in Fig. 4). Others, such as the pair (1, 2) at low frequencies, were rejected due to less-satisfying laser performance, or falloff of laser intensity, e.g., at higher frequencies. As a result, we chose to focus our study on two attractive pairs of transitions: R10.5 (1/2, 3/2) near 1912 cm^{-1} (5 and 6), and R15.5 (1/2, 3/2) near 1926 cm^{-1} (8 and 9).

5. *In situ* NO measurement in laboratory combustion exhaust

The grating scan of the external-cavity laser provided broad wavelength tuning for a complete single-scan survey spectrum in a time of under five seconds. The absorption SNR was improved with faster scanning over shorter wavelength ranges by driving the external-cavity length with a piezoelectric element on the cavity mirror. The piezo element was driven by an external controller (Thorlabs MDT694A) with a sinusoidally varying voltage between 0 and 100 V, which caused the output wavelength of the laser to be varied approximately 2 cm^{-1} at a maximum frequency of 100 Hz. Such a scan range was sufficient for direct-absorption scans of a single NO transition. A solid germanium etalon (free-spectral-range = 0.016 cm^{-1}) was used to characterize the relative variation of laser frequency in such experiments. Validation of the line strength of candidate NO lines was carried out in a room-temperature cell filled with NO mixture at reduced pressure (<100 Torr). The measured line strengths (R10.5 doublets and R15.5 doublets) all agreed

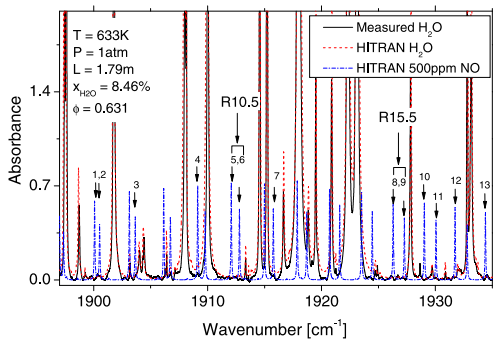

Fig. 4. Solid line shows the H_2O absorption measured at 633 K (680 F) in the effluent of a fuel-lean C_2H_4/air flame. Dashed red line shows the H_2O absorption simulated from HITRAN and the dashed blue line shows the positions of the NO transitions and the absorption predicted by HITRAN for 500 ppm of NO. Arrows show potential transitions and selected windows in the water vapor spectra for NO detection.

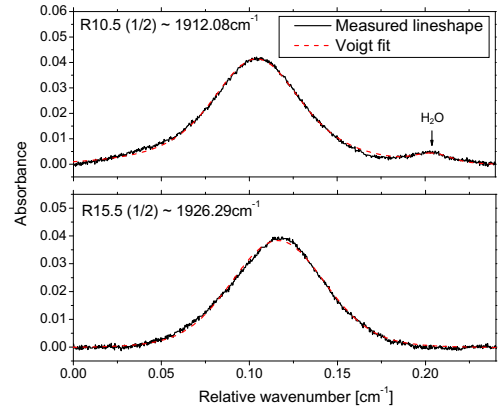

Fig. 5. Measured direct-absorption NO lineshapes in combustion exhaust for fuel-lean C_2H_4/O_2/Ar flame with 22 ppm of added NO at 1 atm, 600 K, $L = 1.79$ m.

with those listed in HITRAN database [18] within 5%, while the declared uncertainty of the latter is reported in the range 5–10%.

To evaluate the sensor performance in hot combustion gases, we performed a series of experiments in the laboratory facility described in Fig. 2. An ethylene flame with mixture of 21% O_2 in Ar provided a NO-free combustion exhaust background gas mixture under atmospheric pressure. A trace amount of NO was seeded into the exhaust to produce precise NO concentrations, which were compared with traditional extractive sampling and a chemiluminescence NO_x analyzer (Teledyne instruments model 200EH). Scanned-wavelength direct absorption with piezo tuning of the laser at 80 Hz was used. The integrated absorbance area as defined in Eq. (2) was calculated from measured spectrally resolved absorption to infer NO concentration.

The measurements were conducted in combustion exhaust gases at 600 K with a flame equivalence ratio $\phi = 0.85$ using four different absorption transitions, R10.5 (1/2), R10.5 (3/2), R15.5 (1/2), and R15.5 (3/2). Figure 5 shows an example of the measured lineshapes of the R10.5 (1/2) and R15.5 (1/2) lines in single 80 Hz scan at the lowest concentration level measured in the above experiment. The R15.5 line was completely free from water interference at this temperature. The R10.5 line had a neighboring water feature about 0.1 cm^{-1} to the blue of the R10.5 line, but that feature was relatively weak and far away enough from the NO line center that a two-line Voigt fit provided accurate absorbance measurements. Note that at atmospheric pressure, the Λ-split doublets are sufficiently broadened and overlapped to be treated as a single line with summed line strength. This measurement of NO concentration of 37.6 ppm m had a SNR of over 70 for both of the two lines, corresponding to a single-sweep absorbance mean-square-root noise level of 5.45×10^{-5}, and a single-sweep detection limit of about 0.5 ppm m. This detection limit was improved where multiple scans were averaged, and the noise level growth was found to be inversely proportional to the square root of averaging time. As a result, a noise-limited detection limit of <60 ppb m (Hz)$^{-1/2}$ was obtained for the piezo-scan direct-absorption measurements of NO at 600 K in water-laden combustion exhaust; this is lower than the detection limit of 200 ppb m (Hz)$^{-1/2}$ measured previously at 630 K in a heated cell using a pulsed QCL by Wysocki et al. [17]. Figure 6 shows the result of measurements for NO concentrations from 20 ppm to 95 ppm using the four selected

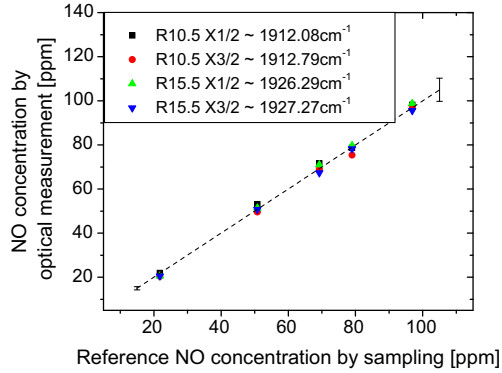

Fig. 6. NO concentrations in combustion exhaust gas measured by absorption spectroscopy versus NO_x analyzer measurements in fuel-lean exhaust of C_2H_4/O_2/Ar flame with added NO at 600 K.

transitions. The measured concentrations consistently agreed with the NO_x analyzer measurements within 5%. A 2σ uncertainty of 5% of the optical measurement was estimated based on the contributions from temperature (2%), pressure (0.5%), gas mixture (3%), and statistical uncertainty of repetitive measurements (2%).

We note that at fuel-rich conditions where significant amounts of CO may be produced (e.g., thousands of ppm), the absorption features in the tail of the P-branch of the CO fundamental band could possibly become a source of interference for measurement of NO in combustion exhaust at elevated temperatures. Among the four lines studied above, the R15.5 (3/2) line near 1927.27 cm^{-1} would be interfered by line P47 of CO near 1927.25 cm^{-1}, whereas the other three remain excellent choices for sensitive NO detection at 600 K. However, in practical cases such as at the economizer exit of a coal-fired power plant where CO level is typically controlled below the 100 ppm level, such interference can be neglected.

6. Real-time NO measurements in coal-fired power plant

The sensor was demonstrated for real-time *in situ* measurement of NO in the exhaust of the economizer of a 300 MW unit of a 1.2 GW pulverized-coal-fired power utility. At the exit of the economizer the temperature was ~624 K (663 F), and access ports were already available to sample an optical path length of 3.05 m (10 feet) across the central portion of the duct. In this power plant environment, the dominant noise sources included mechanical vibration and beam steering in the turbulent flow rather than detector or laser noise. Figure 7 shows a schematic of the sensor setup across the exhaust duct, where a visible green laser beam was used to aid alignment of the mid-IR beam. We were able to achieve >30% transmission of the 0.5 mm diameter laser beam with only 1 in. diameter collection optics. The current experiment used wavelength-scanned direct absorption, providing straightforward quantitative data interpretation without calibration. In the future, we anticipate investigating wavelength modulation spectroscopy, which could provide facile real-time monitoring. Figure 8 shows a representative time-resolved measurement of NO concentration over a period of 60 min with a 5-s time-resolution. During the period shown in Fig. 8, the amount of excess air was first increased, causing a slight rise in NO level. Then the shutdown and recovery of the SNCR system were observed, indicating ~30% changes in NO concentration. When the urea injection ceased, the NO concentration rose to ~120 ppm in the absence of SNCR. Once the urea injection

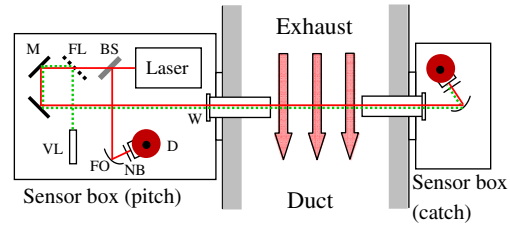

Fig. 7. Sensor setup for field measurement of NO in a coal-fired power plant. Two sensor boxes (pitch and catch sides) are mounted on opposite sides of the duct wall. M, mirror; FL, flipper mirror; BS, beam splitter; VL, visible laser; FO, focusing mirror; NB, narrow band filter; D, detector; W, window. Solid red line: mid-IR laser beam path; dotted green line: visible laser beam path. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this paper.)

was restarted along with the reduction of excess air to normal operation level, the NO dropped immediately, in less than one minute, to an NO concentration of 77 ppm, and then slowly recovered a value near 85 ppm, typical of full-load operation.

The NO_x concentration also was observed far downstream by continuous emission monitoring (CEM) at the exhaust stack with standard sampling and chemiluminescence methods, and these data are shown in the Fig. 8. The optical sensor measured the NO concentration prior to the outlet of the reheat boiler, before any particulate cleanup or other exhaust gas treatment, while the CEM measurement was made significantly further downstream and included exhaust gases from two separate boilers (labeled by convention reheat and superheat). Furthermore, the CEM measurement is of NO_x (i.e., $NO + NO_2$). Hence, even though the urea injection and variation of input air were performed in parallel on both combustors, quantitative agreement of the optical NO measurement to the CEM data is not expected. Note however that the time-variation of the *in situ* NO concentration variation measured by the optical sensor was in consistent agreement with the CEM NO_x concentration. Moreover, the optical measurements provide greatly improved time-resolution with no significant time lag, whereas the CEM had a time-resolution of 1 min and a lag of 6–7 min. In this highly successful demonstration, the sensor operated continuously for a period of eight hours, without failure and without adjustment. The practicality of long term use of such a system in the boiler environment will require replacing the current liquid-nitrogen-cooled detectors with thermoelectrically-cooled versions reducing the cost complexity of the widely tunable EC-QCL and incorporation of an automated optical alignment system. With such refinements, a mid-IR

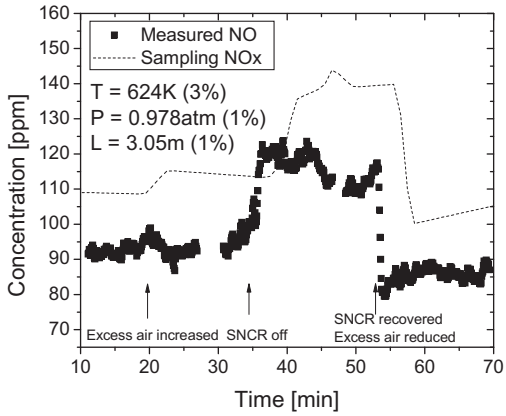

Fig. 8. Optical measurement of NO concentration during the failure and recovery of the SNCR NO_x control system, and comparison with CEM data for NO_x ($NO + NO_2$) recorded in the exhaust stack.

QCL-based NO sensor shows excellent potential for coal-fired applications.

7. Conclusions

Measurements of water vapor absorption were made in combustion gases as a function of temperature ($T < 700$ K) in the R-branch region of the NO fundamental vibrational transition near 5.2 μm using an EC-QCL. The wide tuning range of the laser (1880 cm^{-1} to 1951 cm^{-1}) provided the opportunity to assess the most promising NO transitions for combustion-gas sensing. Single direct-absorption scans across this spectral region, in flame gases at 633 K, allowed confirmation of the high temperature water vapor absorption spectra, and hence the optimum NO transitions. With the R10.5 (1/2, 3/2), R15.5 (1/2, 3/2) transitions, selected as optimum based on relatively low water interference and better laser performance, NO was detected using piezo-scan direct-absorption spectroscopy at 80 Hz in a 1.79 m-long uniform-temperature combustion exhaust rig fed by combustion gases from an ethylene flame. Measurements in combustion gases at 600 K with NO concentrations between 20 ppm and 95 ppm agreed with traditional sampling chemiluminescence NO_x analyzer within 5%. A noise-limited detection limit of <60 ppb m $(Hz)^{-1/2}$ was achieved. The sensor was then applied in a field measurement of NO at the exit of the economizer in a coal-fired power plant, including a 20-min period of shutdown and recovery of the SNCR NO_x reduction system. A 30% change in NO level was measured with 5 s time-resolution, enabling direct, real-time observation of the transient performance of the SNCR system. The sensor met all the targeted performance objectives in this field measurement campaign, providing direct evidence of the potential of this sensor for real-time *in situ* measurement of NO in harsh environments such as coal-fired power plants.

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